Now as σ runs through all elements of $G(K/F)$, we see that $\tau\sigma$ also runs through all elements because $G(K/F)$ is a group. Thus $\overline{\tau}(f(x)) = f(x)$ for each $\tau \in G(K/F)$, so $f(x) \in F[x]$.

b. Because $\sigma(\alpha)$ is a conjugate of α over F for all $\sigma \in G(K/F)$, we see that $f(x)$ has precisely the conjugates of α as zeros. Because $f(\alpha) = 0$, we know by Theorem 29.13 that $p(x) = \text{irr}(\alpha, F)$ divides $f(x)$. Let $f(x) = p(x)q_1(x)$. If $q_1(x) \neq 0$, then it has as zero some conjugate of α whose irreducible polynomial over F is again $p(x)$, so $p(x)$ divides $q_1(x)$ and we have $f(x) = p(x)^2 q_2(x)$. We continue this process until we finally obtain $f(x) = p(x)^r c$ for some $c \in F$. Because $p(x)$ and $f(x)$ are both monic, we must have $f(x) = p(x)^r$.

Now $f(x) = p(x)$ if and only if $\deg(\alpha, F) = |G(K/F)| = [K : F]$. Because $\deg(\alpha, F) = [F(\alpha) : F]$, we see that this occurs if and only if $[F(\alpha) : F] = [K : F]$ so that $[K : F(\alpha)] = 1$ and $K = F(\alpha)$.

25. In the one-to-one correspondence between subgroups of $G(K/F)$ and fields E where $F \leq E \leq K$, the diagram of subgroups of $G(K/F)$ is the inverted diagram of such subfields E of K. Now $E \vee L$ is the smallest subfield containing both E and L, and thus must correspond to the largest subgroup contained in both $G(K/E)$ and $G(K/L)$. Thus

$$G(K/(E \vee L)) = G(K/E) \cap G(K/L).$$

26. Continuing to work with the one-to-one corespondence and diagrams mentioned in the solution to Exercsie 25, we note that $E \cap L$ is the largest subfield of K contained in both E and L. Thus its group must be the smallest subgroup of $G(K/F)$ containing both $G(K/E)$ and $G(K/L)$. Therefore

$$G(K/E \cap L)) = G(K/E) \vee G(K/L),$$

which is defined as the intersection of all subgroups of $G(K/F)$ that contain both $G(K/E)$ and $G(K/L)$, and is called the **join** of the subgroups $G(K/E)$ and $G(K/L)$.

54. Illustrations of Galois Theory

1. Recall that if $x^4 + 1$ has a factorization into polynomials of lower degree in $\mathbb{Q}[x]$, then it has such a factorization in $\mathbb{Z}[x]$; see Theorem 23.11. The polynomial does not have a linear factor, for neither 1 nor -1 are zeros of the polynomial. Suppose that

$$x^4 + 1 = (x^2 + ax + b)(x^2 + cx + d)$$

for $a, b, c, d \in \mathbb{Z}$. Equating coefficients of x^3, x^2, x, and 1 in that order, we find that

$$a + c = 0, \quad ac + b + d = 0, \quad ad + bc = 0, \quad \text{and} \quad bd = 1.$$

If $b = d = 1$, then $ac + 2 = 0$ so $ac = -2$ and $a^2 = 2$, which is impossible for an integer a. The other possibility, $b = d = -1$, leads to $a^2 = -2$ which is also impossible. Thus $x^4 + 1$ is irreducible in $\mathbb{Q}[x]$.

2. The fields corresponding to the subgroups $G(K/\mathbb{Q}), H_2, H_4, H_7$, and $\{\rho_0\}$ are either derived in the text or are obvious. We turn to the other subgroups, H_1, H_3, H_5, and H_8. Both H_1 and H_3 have order 4 and must have fixed fields of degree 2 over $\mathbb{Q}$. Recalling that the fixed field of H_2 is $\mathbb{Q}(i)$, we see that the other two obvious extensions of degree 2, namely $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(i\sqrt{2})$, must be fixed fields of H_1 and H_3. We find that $\delta_1(\sqrt{2}) = \delta_1((\sqrt[4]{2})^2) = (i\sqrt[4]{2})^2 = -\sqrt{2} \neq \sqrt{2}$. Thus H_3, which contains δ_1, must have $\mathbb{Q}(i\sqrt{2})$ as its fixed field, and H_1 must have $\mathbb{Q}(\sqrt{2})$ as its fixed field.

For the fixed field of $H_5 = \{\rho_0, \mu_2\}$ we need to find some elements left fixed by μ_2. Because $\mu_2(\alpha) = -\alpha$ and $\mu_2(i) = -i$, the product $i\alpha$ is an obvious choice. Now $i\sqrt[4]{2}$ is a zero of $x^4 - 2$ which is irreducible, so $\mathbb{Q}(i\sqrt[4]{2})$ is of degree 4 over $\mathbb{Q}$ and left fixed by H_5.

Because ρ_2 leaves i fixed and maps α into $-\alpha$, it leaves i and $\alpha^2 = \sqrt{2}$ fixed. Thus the fixed field of $H_6 = \{\rho_0, \rho_2\}$ is $\mathbb{Q}(i, \sqrt{2})$.

To find an element left fixed by $H_8 = \{\rho_0, \delta_2\}$, we form

$$\beta = \rho_0(\alpha) + \delta_2(\alpha) = \alpha - i\alpha = \sqrt[4]{2}(1 - i).$$

(Note that because H_8 is a group, this sum of elements of H_8 applied to any one element in the field is sure to be left fixed by H_8.) Now $\beta^4 = 2(-4) = -8$, so β is a zero of $x^4 + 8$. This polynomial does not have $\pm 1, \pm 2, \pm 4$, or ± 8 as a zero, so it has no linear factors. If

$$x^4 + 8 = (x^2 + ax + b)(x^2 + cx + d),$$

then $a + c = 0, ac + b + d = 0, ad + bc = 0$, and $bd = 8$. From $a + c = 0$ and $ad + bc = 0$, we find that $ad - ba = 0$ so $a(d - b) = 0$ and either $a = 0$ or $b = d$. Because $bd = 8$, we do not have $b = d$, so $a = 0$. But then $b + d = 0$ so $b = -d$, which again cannot satisfy $bd = 8$. Thus $x^4 + 8$ is irreducible, and $\mathbb{Q}(\sqrt[4]{2}(1 - i))$ has degree 4 over $\mathbb{Q}$ and is left fixed by H_8, so it is the fixed field of H_8.

3. The choices for primitive elements given in the text answers, the corresponding polynomials and the irreducibility of the polynomials are obvious or proved in the text or the preceding solution, except for the first and fourth answers given.

For the case $\mathbb{Q}(\sqrt{2}, i)$, let $\beta = \sqrt[4]{2}$ and $\gamma = i$. The proof of Theorem 51.15 shows that for $a \in \mathbb{Q}, \beta + a\gamma$ is a primitive element if

$$(\beta_i - \beta)/(\gamma - \gamma_j) \neq a$$

where β_i can be any conjugate of β and γ_j is any conjugate other than γ of γ. Now $\gamma - \gamma_j$ is always $i - (-i) = 2i$, and because $\beta = \alpha = \sqrt[4]{2}$ in Table 54.5, it is clear from the table that [(conjugate of α) $- \alpha$]$/(2i)$ is never a nonzero element of $\mathbb{Q}$. Thus we can take $a = 1$, and we find that $\sqrt[4]{2} + i$ is a primitive element. Let $\delta = \sqrt[4]{2} + i$. Then $\delta - i = \sqrt[4]{2}$ so

$$(\delta - i)^4 = \delta^4 - 4\delta^3 i - 6\delta^2 + 4\delta i + 1 = 2$$

so

$$\delta^4 - 6\delta^2 - 1 = (4\delta^3 - 4\delta)i$$

Squaring both sides, we obtain

$$\delta^8 - 12\delta^6 + 34\delta^4 + 12\delta^2 + 1 = -16\delta^6 + 32\delta^4 - 16\delta^2$$

so $\delta^8 + 4\delta^6 + 2\delta^4 + 28\delta^2 + 1 = 0$. Thus δ is a zero of $x^8 + 4x^6 + 2x^4 + 28x^2 + 1 = 0$. Because we know that $\mathbb{Q}(\delta)$ is of degree 8 over $\mathbb{Q}$, this must be $\text{irr}(\mathbb{Q}, \delta)$.

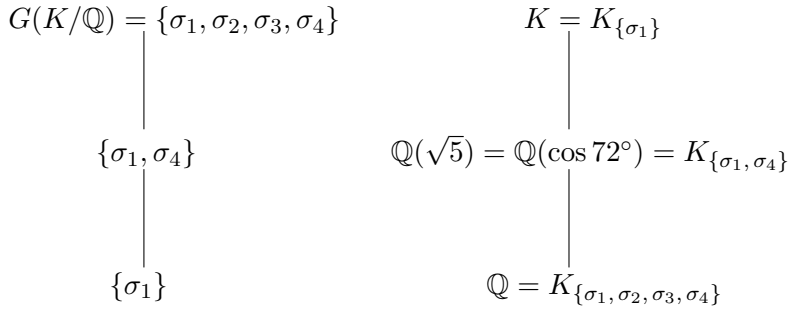
For $\mathbb{Q}(\sqrt{2}, i)$, we have [(conjugate of $\sqrt{2}$) $- \sqrt{2}$]$/2i$ is never a nonzero element of $\mathbb{Q}$, so $\sqrt{2} + i$ is a primitive element. If $\delta = \sqrt{2} + i$, then $\delta - i = \sqrt{2}$ and $\delta^2 - 2\delta i - 1 = 2$. Then $\delta^2 - 3 = 2\delta i$ so $\delta^4 - 6\delta^2 + 9 = -4\delta^2$ and $\delta^4 - 2\delta^2 + 9 = 0$. Thus δ is a zero of $x^4 - 2x^2 + 9$, and because $\mathbb{Q}(\delta)$ is of degree 4 over $\mathbb{Q}$, we see that this polynomial is irreducible.

4. a. If ζ is a primitive 5th root of unity, then $1, \zeta, \zeta^2, \zeta^3$, and ζ^4 are five distinct elements of $\mathbb{Q}(\zeta)$, and $(\zeta^k)^5 = (\zeta^5)^k = 1^k = 1$ shows that these five elements are five zeros of $x^5 = 1$. Thus $x^5 - 1$ splits in $\mathbb{Q}(\zeta)$.

b. We know that $x^5 - 1 = (x-1)\Phi_5(x)$ where $\Phi_5(x) = x^4 + x^3 + x^2 + 1$ is the irreducible (Corollary 23.17) cyclotomic polyonomial having ζ as a root. Every automorphism of $K = \mathbb{Q}(\zeta)$ over $\mathbb{Q}$ must map ζ into one of the four roots $\zeta, \zeta^2, \zeta^3, \zeta^4$ of this polynomial.

c. Let $\sigma_j \in G(K/\mathbb{Q})$ be the automorphism such that $\sigma(\zeta) = \zeta^j$ for $j = 1, 2, 3, 4$. Then $(\sigma_j\sigma_k)(\zeta) = \sigma_j(\zeta^k) = (\zeta^j)^k = \zeta^{jk} = \sigma_m(\zeta)$ where m is the product of j and k in $\mathbb{Z}_5$. Thus $G(K/\mathbb{Q})$ is isomorphic to the group $\{1, 2, 3, 4\}$ of nonzero elements of $\mathbb{Z}_5$ under multiplication. It is cyclic of order 4, generated by σ_2.

d.



Subgroup diagram Subfield diagram

To find $K_{\{\sigma_1, \sigma_4\}}$, note that $\zeta = \cos 72^\circ + i \sin 72^\circ$ and that $\zeta^4 = \cos(-72^\circ) + i \sin(-72^\circ) = \cos(72^\circ) - i \sin(72^\circ)$. Therefore $\alpha = \sigma_1(\zeta) + \sigma_4(\zeta) = \zeta + \zeta^4 = 2 \cos 72^\circ$ is left fixed by σ_1 and σ_4. Alternatively, doing a bit of computation, we find that

$$\begin{aligned} \alpha^2 &= (\zeta + \zeta^4)^2 = \zeta^2 + 2 + \zeta^3, \\ \alpha &= \zeta + \zeta^4. \end{aligned}$$

Now ζ is a zero of $\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$, so we see that $\alpha^2 + \alpha - 1 = 0$, so α is a zero of $x^2 + x - 1$ which has zeros $(-1 \pm \sqrt{5})/2$. Thus we can also describe $\mathbb{Q}(\alpha)$ as $\mathbb{Q}(\sqrt{5})$.

5. The splitting field of $x^5 - 2$ over $\mathbb{Q}(\zeta)$ is $\mathbb{Q}(\zeta, \sqrt[5]{2})$, because $\sqrt[5]{2}, \zeta\sqrt[5]{2}, \zeta^2\sqrt[5]{2}, \zeta^3\sqrt[5]{2}$, and $\zeta^4\sqrt[5]{2}$ are the five zeros of $x^5 - 2$. The Galois group $\{\sigma_0, \sigma_1, \sigma_2, \sigma_3, \sigma_4\}$ is described by the table.

	σ_0	σ_1	σ_2	σ_3	σ_4
$\sqrt[5]{2} \to$	$\sqrt[5]{2}$	$\zeta\sqrt[5]{2}$	$\zeta^2\sqrt[5]{2}$	$\zeta^3\sqrt[5]{2}$	$\zeta^4\sqrt[5]{2}$

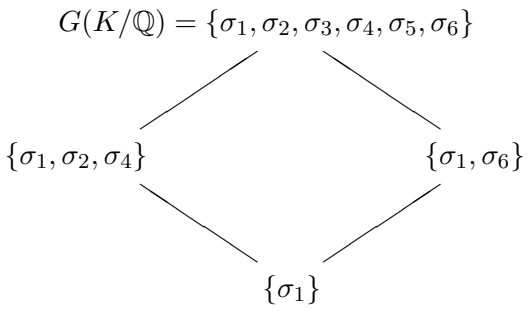
We have $(\sigma_j\sigma_k)(\sqrt[5]{2}) = \sigma_j(\zeta^k(\sqrt[5]{2})) = \zeta^k\zeta^j(\sqrt[5]{2}) = \zeta^{j+k}(\sqrt[5]{2})$ from which we see that the Galois group is isomorphic to $\langle \mathbb{Z}_5, + \rangle$, so it is cyclic of order 5.

6. a. If ζ is a primitive 7th root of unity, then $1, \zeta, \zeta^2, \zeta^3, \zeta^4, \zeta^5$, and ζ^6 are seven distinct elements of $\mathbb{Q}(\zeta)$, and $(\zeta^k)^7 = (\zeta^7)^k = 1^k = 1$ shows that these seven elements are seven zeros of $x^7 = 1$. Thus $x^7 - 1$ splits in $\mathbb{Q}(\zeta)$.

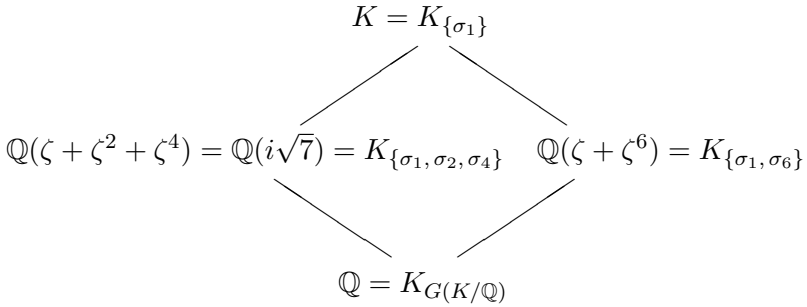
b. We know that $x^7 - 1 = (x-1)\Phi_7(x)$ where $\Phi_7(x) = x^6 + x^5 + x^4 + x^3 + x^2 + 1$ is the irreducible (Corollary 23.17) cyclotomic polyonomial having ζ as a root. Every automorphism of $K = \mathbb{Q}(\zeta)$ over $\mathbb{Q}$ must map ζ into one of the six roots $\zeta, \zeta^2, \zeta^3, \zeta^4, \zeta^5, \zeta^6$ of this polynomial.

c. Let $\sigma_j \in G(K/\mathbb{Q})$ be the automorphism such that $\sigma(\zeta) = \zeta^j$ for $j = 1, 2, 3, 4, 5, 6$. Then $(\sigma_j\sigma_k)(\zeta) = \sigma_j(\zeta^k) = (\zeta^j)^k = \zeta^{jk} = \sigma_m(\zeta)$ where m is the product jk in $\mathbb{Z}_7$. Thus $G(K/\mathbb{Q})$ is isomorphic to the group $\{1, 2, 3, 4, 5, 6\}$ of nonzero elements of $\mathbb{Z}_7$ under multiplication. It is cyclic of order 6, generated by σ_3.

d.



Subgroup diagram



Subfield diagram

Clearly $\alpha = \zeta + \zeta^2 + \zeta^4$ is left fixed by $\{\sigma_1, \sigma_2, \sigma_4\}$. Computing, we find that

$$\begin{aligned} \alpha^2 &= \zeta^2 + \zeta^4 + \zeta + 2\zeta^3 + 2\zeta^6 + 2\zeta^5, \\ \alpha &= \zeta + \zeta^2 + \zeta^4. \end{aligned}$$

Thus we find that $\alpha^2 + \alpha = 2(\zeta^6 + \zeta^5 + \zeta^4 + \zeta^3 + \zeta^2 + \zeta)$. Because ζ is a zero of

$$\Phi_7(x) = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1,$$

we see at once that $\alpha^2 + \alpha + 2 = 0$. The zeros of $x^2 + x + 2$ are $(-1 \pm i\sqrt{7})/2$, and we see that $\mathbb{Q}(\alpha) = \mathbb{Q}(i\sqrt{7})$.

Working in an analogous way for the subgroup $\{\sigma_1, \sigma_6\}$, we form the element $\beta = \zeta + \zeta^6$ which is left fixed by this subgroup. Computing, we find that

$$\begin{aligned} \beta^3 &= (\zeta + \zeta^6)^3 = \zeta^3 + 3\zeta + 3\zeta^6 + \zeta^4, \\ \beta^2 &= (\zeta + \zeta^6)^2 = \zeta^2 + 2 + \zeta^5, \\ \beta &= \zeta + \zeta^6. \end{aligned}$$

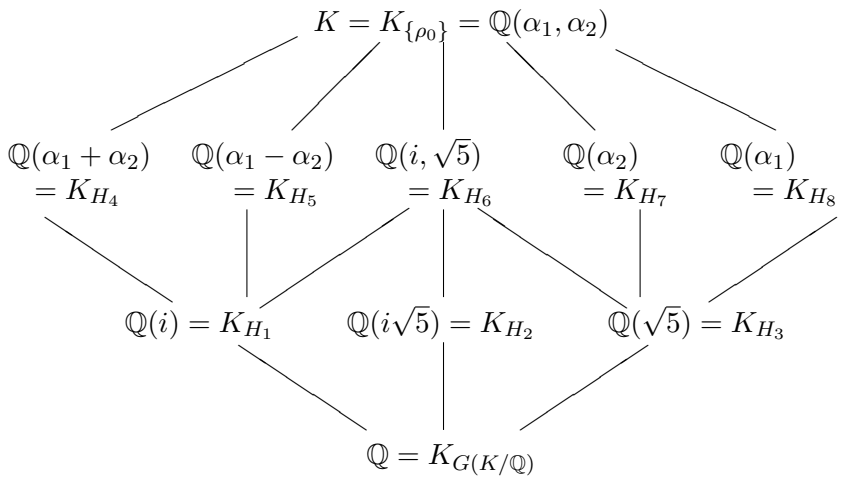
Recalling that $\Phi_7(\zeta) = 0$ as above, we find that $\beta^3 + \beta^2 - 2\beta - 1 = 0$. Thus β is a zero of $x^3 + x^2 - 2x + 1$ which is irreducible because it has no zero in $\mathbb{Z}$.

7. Now $x^8-1=(x^4+1)(x^2+1)(x-1)(x+1)$. Example 54.7 shows that the splitting field of x^4+1 contains i, which is a zero of x^2+1. Thus the splitting field of x^8-1 is the same as the splitting field of x^4+1, whose group was completely described in Example 54.7. This is the "easiest way possible" for us to describe this group.

8. Using the quadratic formula to find α such that $\alpha^4-4\alpha^2-1=0$, we find that $\alpha^2=(4\pm\sqrt{20})/2=2\pm\sqrt{5}$ so the possible values for α are $\pm\sqrt{2+\sqrt{5}}$ and $\pm i\sqrt{\sqrt{5}-2}$. We see that the splitting field of $x^4-4x^2-1=0$ can be generated by adjoining in succession $\sqrt{5}$, $\sqrt{\sqrt{5}+2}$, and $\sqrt{2-\sqrt{5}}$. Thus it has degree $2^3=8$ over $\mathbb{Q}$. It can obviously be generated by adjoining $\alpha_1=\sqrt{\sqrt{5}+2}$ and $\alpha_2=i\sqrt{\sqrt{5}-2}$. Let $K=\mathbb{Q}(\alpha_1,\alpha_2)$. The eight elements of $G(K/\mathbb{Q})$ are given by this table.

	ρ_0	ρ_1	ρ_2	ρ_3	μ_1	μ_2	δ_1	δ_2
$\alpha_1\to$	α_1	α_2	$-\alpha_1$	$-\alpha_2$	α_2	$-\alpha_2$	$-\alpha_1$	α_1
$\alpha_2\to$	α_2	-α_1	$-\alpha_2$	α_1	α_1	$-\alpha_1$	α_2	$-\alpha_2$

The group is isomorphic to D_4 and the notation here is chosen to coincide with the notation used in Example 8.10. The subgroup diagram is identical with that in Fig. 54.6(a). Here is the subfield diagram.



Subfield diagram

We now check most of this diagram. Note that α_1^2 is left fixed by $H_3=\{\rho_0,\rho_2,\delta_1,\delta_2\}$ and that $\alpha_1^2=\sqrt{5}+2$, so the fixed field of H_3 is $\mathbb{Q}(\sqrt{5})$. Note also that $\alpha_1\alpha_2=i$ is left fixed by $H_1=\{\rho_0,\rho_2,\mu_1,\mu_2\}$ so the fixed field of H_1 is $\mathbb{Q}(i)$. Then $H_6=H_1\cap H_3$ leaves both i and $\sqrt{5}$ fixed. Also K_{H_2} must be the only remaining extension of $\mathbb{Q}$ of degree 2, and $\mathbb{Q}(i\sqrt{5})$ fits the bill.

The remaining fields are trivial to check because they are described in terms of α_1 and α_2. For example, to see that $\mathbb{Q}(\alpha_1)$ is the fixed field of H_8, we need only note that $\{\rho_0,\delta_2\}=H_8$ is the set of elements leaving α_1 fixed in the action table shown earlier.

9. We develop some formulas to use in this exercise and the next one. For simple notation, we denote symmetric expressions in the indeterminates y_1,y_2,y_3 by the notation S(formula) where the formula indicates the nature of *one* summand of the expression. Thus we write

$$\begin{aligned} s_1 &= y_1+y_2+y_3=S(y_i),\\ s_2 &= y_1y_2+y_1y_3+y_2y_3=S(y_iy_j),\\ s_3 &= y_1y_2y_3=S(y_iy_jy_k). \end{aligned}$$

Note that the subscripts i, j, k do not run independently through values from 1 to 3; we always have $i \neq j \neq k$. The formula simply indicates the nature of one, typical term in the symmetric expression.

a. We now express some other symmetric expression in terms of s_1, s_2, and s_3. Theorem 54.2 asserts that this is possible. If you write them out for subscript values from 1 to 3, you can see why they hold. Equation (1) is the answer to Part(**a**).

$$y_1^2 + y_2^2 + y_3^2 = S(y_i^2) = [S(y_i)]^2 - 2S(y_i y_j) = s_1^2 - 2s_2. \tag{1}$$

b. We have

$$S(y_i^2 y_j) = S(y_i y_j)S(y_i) - 3S(y_i y_j y_k) = s_1 s_2 - 3s_3. \tag{2}$$

We can now get the answer to Part(**b**), using formula (2),

$$\frac{y_1}{y_2} + \frac{y_2}{y_1} + \frac{y_1}{y_3} + \frac{y_3}{y_1} + \frac{y_2}{y_3} + \frac{y_3}{y_2} = S\left(\frac{y_i}{y_j}\right) =$$

$$\begin{aligned} \frac{y_1^2 y_3 + y_2^2 y_3 + y_1^2 y_2 + y_3^2 y_2 + y_2^2 y_1 + y_3^2 y_1}{y_1 y_2 y_3} &= \frac{S(y_i^2 y_j)}{y_1 y_2 y_3} \\ &= \frac{s_1 s_2 - 3s_3}{s_3}. \end{aligned}$$

10. We have $x^3 - 4x^2 + 6x - 2 = (x-\alpha_1)(x-\alpha_2)(x-\alpha_3)$. Therefore the elementary symmetric expressions in α_1, α_2, and α_3 are given by

$$\begin{aligned} s_1 &= \alpha_1 + \alpha_2 + \alpha_3 = 4, \\ s_2 &= \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3 = 6, \\ s_3 &= \alpha_1\alpha_2\alpha_3 = 2. \end{aligned}$$

We feel free to make use of the formulas (1) and (2) of the solution to the preceding exercise, and using the notation there, we also have the relation

$$S(y_i^2 y_j^2) = [S(y_i y_j)]^2 - 2S(y_i^2 y_j y_k) = s_2^2 - 2s_1 s_3. \tag{3}$$

a. x - 4

b. Let $x^3 + b_2x^2 + b_1x + b_0 = (x - \alpha_1^2)(x - \alpha_2^2)(x - \alpha_3^2)$. Now $b_2 = -\alpha_1^2 - \alpha_2^2 - \alpha_3^2 = -(s_1^2 - 2s_2)$ by formula (1). Evaluating with the values of s_1, s_2, and s_3, we find that $b_2 = -(16 - 12) = -4$. Also $b_1 = \alpha_1^2\alpha_2^2 + \alpha_1^2\alpha_3^2 + \alpha_2^2\alpha_3^2 = s_2^2 - 2s_1s_3$ by formula (3). Evaluating, we find that $b_1 = 36 - 16 = 20$. Finally, $b_0 = -\alpha_1^2\alpha_2^2\alpha_3^2 = -s_3^2 = -4$. Thus the answer is

$$x^3 - 4x^2 + 20x - 4.$$

11. By Cayley's Theorem, every finite group G is isomorphic to a subgroup of S_n where n is the order of G. Now Theorem 54.2 shows that for each positive integer n, there exists a normal extension K of a field E such that $G(K/E) \simeq S_n$. If H is a subgroup of $G(K/E)$ isomorphic to G, then H is the Galois group of K over K_H, where $L = K_H$ is the fixed field of H. Thus H is isomorphic to G and is the Galois group $G(K/L)$ of K over L.

12. **a.** If $\Delta(f) = 0$, then $\alpha_i = \alpha_j$ for some $i \neq j$. Thus $\text{irr}(\alpha_i, F) = \text{irr}(\alpha_j, F)$. Because the irreducible factors of $f(x)$ are all separable and do not have zeros of multiplicity greater than 1, we see that $f(x)$ must have $\text{irr}(\alpha_i, F)^2$ as a factor.

b. Clearly $[\Delta(f)]^2$ is a symmetric expression in the α_i, and hence left fixed by any permutation of the α_i, and thus is invariant under $G(K/F)$. Therefore $[\Delta(f)]^2$ is in F.

c. Consider the effect of a transposition (α_i, α_j) on $\Delta(f)$; it is no loss of generality to suppose $i < j$. The factor $\alpha_i - \alpha_j$ is carried into $\alpha_j - \alpha_i$, so it changes sign. For $k > j, \alpha_k - \alpha_j$ and $\alpha_k - \alpha_i$ for $k > j$ are carried into each other, so they do not contribute a sign change. The same is true of $\alpha_i - \alpha_k$ and $\alpha_j - \alpha_k$ for $k < i$. For $i < k < j$, the terms $\alpha_k - \alpha_i$ and $\alpha_j - \alpha_k$ are carried into $\alpha_k - \alpha_j = -(\alpha_j - \alpha_k)$ and into $\alpha_i - \alpha_k = -(\alpha_k - \alpha_i)$, so they contribute two sign changes. Thus the transposition contributes $1+2(j-i-1)$ sign changes, which is an odd number, and carries $\Delta(f)$ into -$\Delta(f)$. Thus a permutation leaves $\Delta(f)$ fixed if and only if it can be expressed as a product of an even number of transpositions, that is, if and only if it is in $\overline{A}_n$. Hence $G(K/F) \subseteq \overline{A}_n$ if and only if it leaves $\Delta(f)$ fixed, that is, if and only if $\Delta(f) \in F$.

13. Let α and β be algebraic integers and let K be the splitting field of $\text{irr}(\alpha, \mathbb{Q}) \cdot \text{irr}(\beta, \mathbb{Q})$. Now

$$g(x) = \prod_{\sigma \in G(K/\mathbb{Q})} (x - \sigma(\alpha))$$

is a power of $\text{irr}(\alpha, \mathbb{Q})$, and thus has integer coefficients and leading coefficient 1 because α is an algebraic integer. The same is true of

$$h(x) = \prod_{\mu \in G(K/\mathbb{Q})} (x - \mu(\beta)).$$

Now

$$\begin{aligned} k(x) &= \prod_{\sigma,\, \mu \in G(K/\mathbb{Q})} [x - (\sigma(\alpha) + \mu(\beta))] \\ &= \prod_{\sigma \in G(K/\mathbb{Q})} \left[\prod_{\mu \in G(K/\mathbb{Q})} [(x - \sigma(\alpha)) - \mu(\beta)] \right] \\ &= \prod_{\sigma \in G(K/\mathbb{Q})} h(x - \sigma(\alpha)). \end{aligned}$$

Because $h(x)$ has integer coefficients, $h(x - \sigma(\alpha))$ is a polynomial in $x - \sigma(\alpha)$ with integer coefficients. We can view $k(x)$ as a symmetric expression in α and its conjugates over the field $\mathbb{Q}$ involving only integers in $\mathbb{Q}$. By Theorem 54.2, the symmetric expression in α and its conjugates can be expressed as polynomials in the elementary symmetric functions of α and its conjugates, that is, in terms of the coefficients of $g(x)$ or their negatives. Thus $k(x)$ has integer coefficients. Now one zero of $k(x)$ is $\alpha + \beta$, corresponding to the factor where σ and μ are both the identity permutation. Thus $\text{irr}(\alpha + \beta, \mathbb{Q})$ is a factor of the monic polynomial $k(x)$. Because a factorization in $\mathbb{Q}[x]$ can always be implemented in $\mathbb{Z}[x]$ by Theorem 23.11, we see that $\text{irr}(\alpha + \beta, \mathbb{Q})$ is monic with integer coefficients, and hence $\alpha + \beta$ is an algebraic integer. If α is a zero of $f(x)$, then $-\alpha$ is a zero of $f(-x)$ which again has integer coefficients and is monic, so $-\alpha$ is again an algebraic integer.

One can argue that $\alpha\beta$ is an algebraic integer by the same technique that we used for $\alpha + \beta$, considering

$$\prod_{\sigma,\, \mu \in G(K/\mathbb{Q})} [x - \sigma(\alpha)\mu(\beta)] = \prod_{\sigma \in G(K/\mathbb{Q})} \left[\prod_{\mu \in G(K/\mathbb{Q})} [x - \sigma(\alpha)\mu(\beta)] \right].$$